

DEPARTMENT OF MATHEMATICS, IIT - GUWAHATI

Deferred Semester of the Academic Year 2023-2024

MA 101 Mathematics I

Problem Sheet 4: Multiple integrals and applications, Transformation of coordinates.

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1. Evaluate the integrals:

(a) $\int_0^3 \int_0^1 \sqrt{x+y} dx dy$

(b) $\int_R \int \frac{xy^2}{x^2+1} dA, R = \{(x,y) | 0 \leq x \leq 1, -3 \leq y \leq 3\}$

(c) $\int_R \int x \sin(x+y) dA, R = \left[0, \frac{\pi}{6}\right] \times \left[0, \frac{\pi}{3}\right]$.

2. Find the volume of the solid lying under the plane $z = 2x + 5y + 1$ and above the rectangle $\{(x,y) | -1 \leq x \leq 1, 1 \leq y \leq 4\}$.

3. Find the volume of the solid lying under the elliptic paraboloid $\frac{x^2}{4} + \frac{y^2}{9} + z = 1$ and above the square $R = [-1, 1] \times [-2, 2]$.

4. Evaluate the iterated integrals:

(a) $\int_0^1 \int_y^{e^y} \sqrt{x} dx dy$

(b) $\int_0^{\frac{\pi}{2}} \int_0^{\cos \theta} e^{\sin \theta} dr d\theta$.

5. Evaluate the double integrals:

(a) $\int_D \int \frac{2y}{x^2+1} dA, R = \{(x,y) | 0 \leq x \leq 1, 0 \leq y \leq \sqrt{x}\}$

(b) $\int_D \int x \cos y dA, D$ is bounded by $y = 0, y = x^2, x = 1$

(c) $\int_D \int y^3 dA, D$ is the triangular region with vertices (0,2), (1,1) and (3,2).

6. Find the volume of the given solids.

(a) Under the paraboloid $z = x^2 + y^2$ and above the region bounded by $y = x^2$ and $x = y^2$

(b) Under the surface $z = xy$ and above the triangle with vertices (1,1), (4,1) and (1,2)

(c) Bounded by the cylinder $x^2 + z^2 = 9$ and the planes $x = 0, y = 0, z = 0, x + 2y = 2$ in the first octant

(d) Bounded by the planes $x = 0, y = 0, z = 0$, and $x + y + z = 1$.

7. Get an upper bound and lower bound of each of the integrals given below by using the result that if m, M are such that $m \leq f(x,y) \leq M$ for all $(x,y) \in D$ then $m \times A(D) \leq \int_D \int f(x,y) dA \leq M \times A(D)$, where $A(D)$ is the area of D .

(a) $\int_D \int \sqrt{x^3 + y^3} dA, D = [0, 1] \times [0, 1]$

(b) $\int_D \int e^{x^2+y^2} dA, D$ being the disk with center at origin and radius 0.5.

8. Using polar coordinates, find:

- (a) $\iint_R xy dA$, where R is the region in the first quadrant that lies between the circles $x^2 + y^2 = 4$ and $x^2 + y^2 = 25$
- (b) $\iint_D (x^2 + y^2) dA$, where D is the region bounded by the spirals $r = \theta$ and $r = 2\theta$ for $0 \leq \theta \leq 2\pi$
- (c) The volume of a sphere of radius a
- (d) The volume inside both the cylinder $x^2 + y^2 = 4$ and the ellipsoid $4x^2 + 4y^2 + z^2 = 64$
- (e) $\int_0^1 \int_0^{\sqrt{1-x^2}} e^{x^2+y^2} dy dx$
- (f) $\int_0^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} x^2 y^2 dx dy$.

9. Find the mass, center of mass and the moments of inertia I_x , I_y and I_o of the lamina D bounded by the parabola $x = y^2$ and the line $y = x - 2$; the density is $\rho(x, y) = 3$.

10. Find the area of the surface:

- (a) The part of the plane $3x + 2y + z = 6$ that lies in the first octant
- (b) The part of the hyperbolic paraboloid $z = y^2 - x^2$ that lies between the cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$
- (c) The part of the ellipse cut from the plane $z = 2x + 2y + 1$ by the cylinder $x^2 + y^2 = 1$
- (d) The part cut from the paraboloid $z = r^2$ by the cylinder $r = 1$.

11. Evaluate the following triple integrals.

- (a) $\iiint_E 2x dV$, where $E = \{(x, y, z) | 0 \leq y \leq 2, 0 \leq x \leq \sqrt{4 - y^2}, 0 \leq z \leq y\}$
- (b) $\iiint_E x dV$, where E is bounded by the paraboloid $x = 4y^2 + 4z^2$ and the plane $x = 4$.

12. Use triple integral to find the volume of the tetrahedron bounded by the coordinate planes and the plane $2x + 3y + 6z = 12$.

13. Use cylindrical or spherical coordinates whichever is appropriate, to:

- (a) Evaluate $\iiint_E \sqrt{x^2 + y^2} dV$, where E is the region that lies inside the cylinder $x^2 + y^2 = 16$ and between the planes $z = -5$ and $z = 4$
- (b) Evaluate $\iiint_E x^2 dV$, where E is the solid that lies within the cylinder $x^2 + y^2 = 1$, above the plane $z = 0$, and below the cone $z^2 = 4x^2 + 4y^2$
- (c) Evaluate $\iiint_E \sqrt{x^2 + y^2 + z^2} dV$, where E is bounded below by the cone $\phi = \frac{\pi}{6}$ and above by the sphere $\rho = 2$.

14. Find the volume and centroid of the solid E that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = 1$.

15. Show that $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \sqrt{x^2 + y^2 + z^2} e^{-(x^2+y^2+z^2)} dx dy dz = 2\pi$.

16. Use the given transformation to evaluate the integral:

- (a) $\iint_R (3x + 4y) dA$, where R is the region bounded by the lines $y = x$, $y = x - 2$, $y = -2x$ and $y = 3 - 2x$; $x = \frac{1}{3}(u + v)$, $y = \frac{1}{3}(v - 2u)$
- (b) $\iint_R xy dA$, where R is the region in the first quadrant bounded by the lines $y = x$ and $y = 3x$ and the hyperbolas $xy = 1$, $xy = 3$; $x = \frac{u}{v}$, $y = v$.

17. Evaluate the following integrals by making appropriate change of variables:

- (a) $\iint_R \cos\left(\frac{y-x}{y+x}\right) dA$, where R is the trapezoidal region with vertices $(1, 0)$, $(2, 0)$, $(0, 2)$ and $(0, 1)$
- (b) $\iint_R \frac{1}{(x^2 + y^2)^2} dx dy$, where R is the first quadrant region bounded by the circles $x^2 + y^2 = 2x$, $x^2 + y^2 = 6x$ and the circles $x^2 + y^2 = 2y$, $x^2 + y^2 = 8y$.

18. Let R be the solid ellipsoid with constant density δ and boundary surface

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

Use appropriate transformations to show that the mass M of R is $\frac{4}{3}\pi\delta abc$.